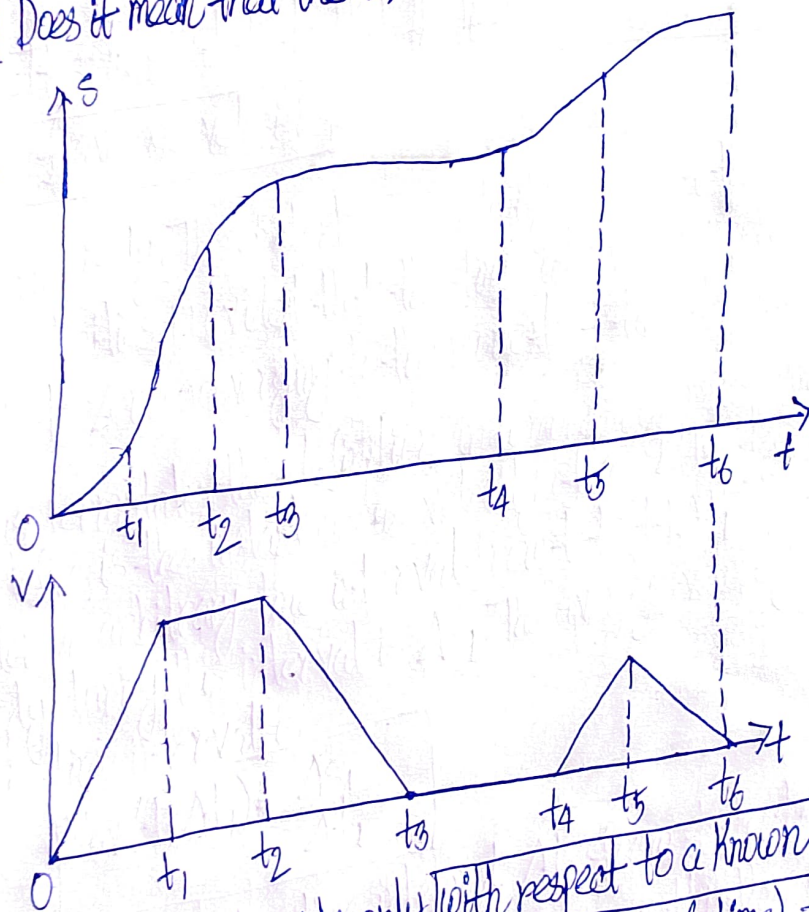


Author: You are absolutely right. I must congratulate you: for the first time in your life you have carried out the so-called operation of differentiation. In other words, you have determined for a given function $s(t)$ its derivative, that is, the function of $v(t)$.

Reader: Does it mean that the instantaneous velocity is a derivative?

Author:



Note that a derivative exists only (with respect to a known initial function). If the initial function is $s(t)$ (path as a function of time), the derivative is the instantaneous velocity.

Let us return now to the graph $s(t)$ shown in Fig. 31. Our previous arguments, and, in particular, relation (4), allow us to transform the graph $s(t)$ into a graph of the derivative, that is, the function $v(t)$. A comparison of the two graphs is given in Fig. 35. I would recommend that you carefully analyze Fig. 35, interpreting it as a comparison of the graph of a function $s(t)$ and the graph of its rate of change.